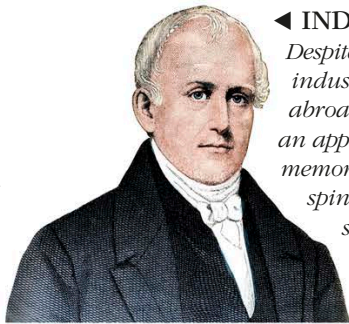




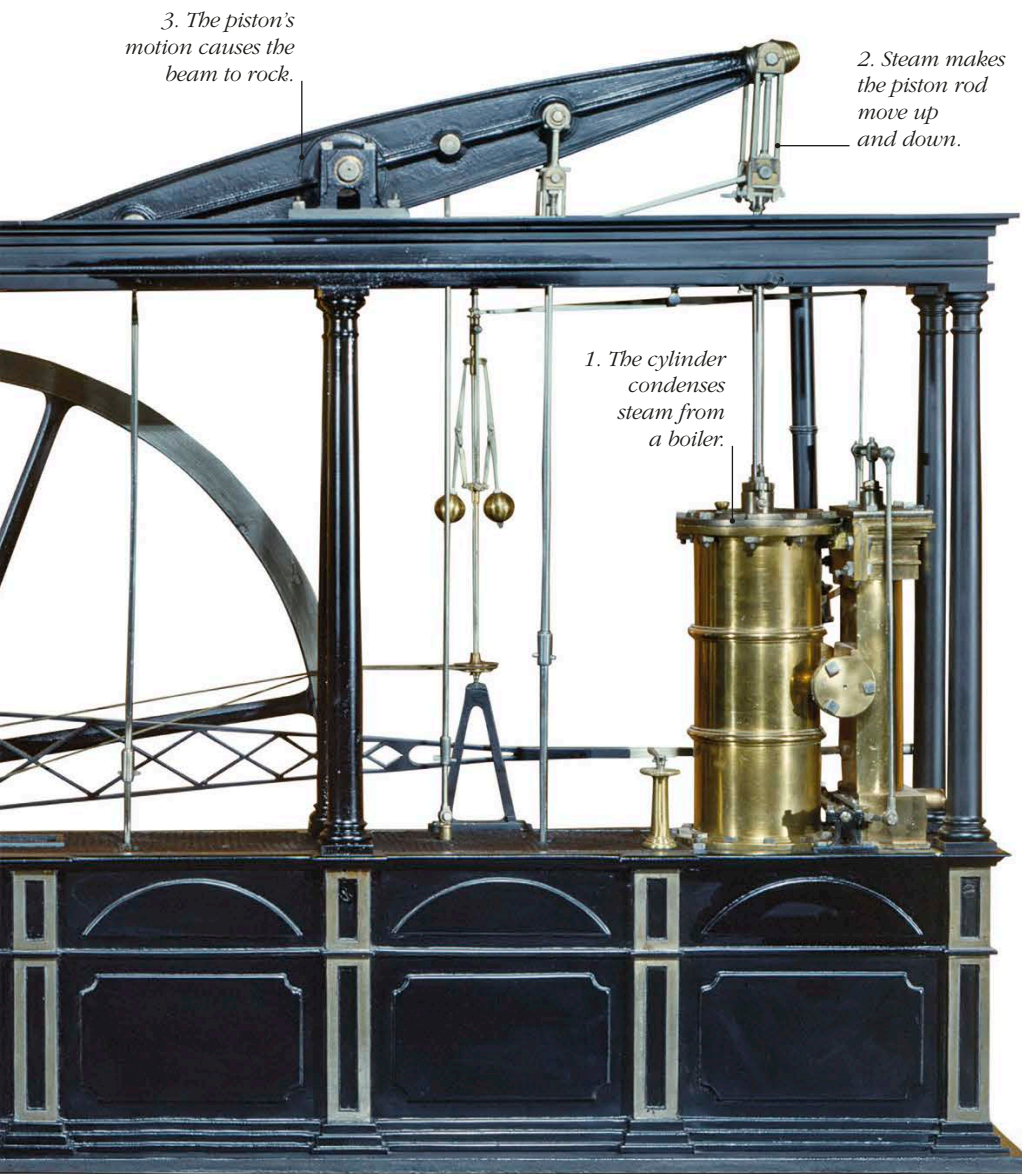
### ▲ THE IRON BRIDGE

In the 18th century, engineers discovered ways to make iron tougher as well as easier to work with. It could then be used to construct buildings and other structures. In 1779, the world's first cast iron bridge (above) was built across the Severn River in Shropshire, England. Demand for high-quality iron soared with the success of the bridge.



### ◀ INDUSTRIAL SECRETS

Despite a British law banning industrial designs being taken abroad, Samuel Slater (left), an apprentice at a cotton mill, memorized the plans of an early spinning machine before sailing to the US in 1789. He used his knowledge to kick-start the US's Industrial Revolution.



3. The piston's motion causes the beam to rock.

2. Steam makes the piston rod move up and down.

1. The cylinder condenses steam from a boiler.

## THE STORY OF STEAM

1712

British engineer Thomas Newcomen builds the first engine able to harness the power of steam. It is used for pumping water out of mines.

1765

Scottish inventor James Watts improves Newcomen's design by condensing steam, which makes the engine far more efficient.

1769

Nicolas-Joseph Cugnot, a French military engineer, designs a steam-driven, self-propelled road vehicle for carrying cannons.

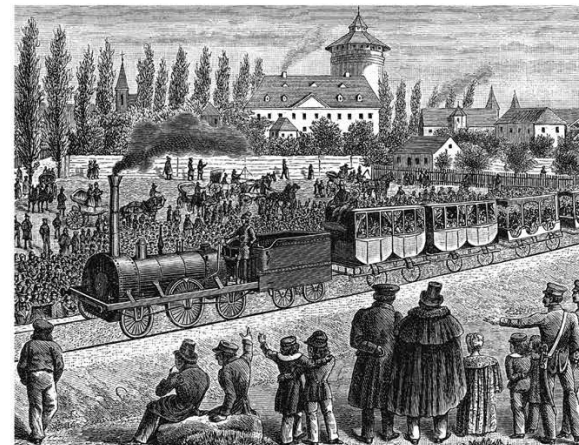
1804

British engineer Richard Trevithick mounts a small steam engine on wheels to create the world's first steam locomotive.

1807

In the US, Robert Fulton builds the first commercial steamboat to carry passengers on the Hudson River.

1750-1900



### ▲ RAILROAD BOOM

In 1830, the world's first railroad running services solely powered by steam locomotives opened in northwestern England. Five years later, the Ludwigs-Eisenbahn company opened the first steam-operated train line in modern-day Germany (above). By the 1840s, railroad construction was booming all over Europe, making long-distance travel quicker and easier.